

Bo Hu

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EDUCATION

New York University, Tandon School of Engineering *Brooklyn, NY* May 2027 (expected)
B.S., Mathematics & Computer Science · GPA: 3.5/4.0 · Exchange: NYU Abu Dhabi (Fall 2025)
Coursework: Algorithms, Operating Systems, UNIX System Programming, Differential Equations, PDEs, Probability, Real Analysis

Stanford University, Summer Session *Palo Alto, CA* Jun–Aug 2022
GPA: 3.8/4.0 (8 credits) · *Coursework:* Data Structures, Astrophysics

EXPERIENCE

Teleoperation Systems, NYU Robotics Club *Brooklyn, NY* Feb 2026–Present

- Contributing to NYU’s HSRN datacenter teleoperation project, developing remote robot control and video feedback systems under *ROS 2 Humble*.
- Developing a real-time video streaming pipeline with the *Viam* C++ SDK and *OpenCV*; benchmarking latency across resolutions and frame rates, and investigating transmission efficiency optimizations.

Motion Team Lead, NYU RoboSub *Brooklyn, NY* Jan 2024–Present

- Implemented and tuned PID controllers using IMU and depth sensor feedback for 6-DOF underwater trajectory tracking; deployed on Jetson with *rosserial* for MCU communication and sensor integration.
- Designed autonomy logic and integrated *YOLO* perception models with the control pipeline to complete competition tasks such as gate passage, torpedo deployment, and path following.
- Ported the control stack to *Gazebo 11* under *ROS Noetic*, leveraging *Project DAVE*’s hydrodynamic and thruster physics; authored custom URDF with mesh geometries and simulation world files for competition environments.

Member, NYU OSIRIS Lab *Brooklyn, NY* Oct 2024–May 2025

- Leads the Side-Channel Attack project; completed the OSIRIS Recruit CTF and participates in weekly CTF scrimms.

IT Operations Intern, Shanghai Data Exchange *Shanghai, China* Jun 2024–Aug 2024

- Automated security audit report generation with a cron-scheduled *Selenium* and Baidu OCR workflow that handles login, screenshots, text extraction, and template filling.
- Set up Smokeping probes for network monitoring and built a *Flask*/MySQL dashboard to track TCP processes across servers for incident triage.

Data Analyst Intern, Neumetry Medical Inc. *Fremont, CA* Jul 2021–Aug 2022

- Wrote Python scripts to run Fourier analysis on physiological time-series data, extracting dominant frequency peaks and generating plots for a wearable cardiovascular study.

PROJECTS

Sysmon: Desktop system monitor built with *Tauri*, *React/TypeScript* frontend and *Rust* backend. The backend manages a shared `sysinfo` instance behind a `Mutex` and exposes per-core CPU usage, memory, disk I/O, and local IP to the frontend through *Tauri* IPC commands using `serde` serialization.

Side-Channel Attack Capture Platform, NYU OSIRIS Lab: Built a differential power analysis capture board with an INA333 instrumentation amplifier, ADS7883 ADC, and ESP32C6. The ESP32 reads the ADC over SPI and uses an I2C trigger line to sync captures to AES-128 encryption boundaries on an STM32. Runs on-device correlation power analysis to recover key bytes.

Finance & Economics Time-Series Analysis, NYU Data Science Bootcamp: Modeled macroeconomic time-series data using ARIMA and regularized regression (Lasso, Ridge); performed PCA and correlation analysis across daily, monthly, and quarterly granularities.

SKILLS

Languages: C/C++, Python, TypeScript, SQL, Ruby, Rust
Robotics/Systems: ROS Noetic, ROS 2 Humble, Gazebo, Matlab, Viam, WebRTC, *rosserial*, Linux/UNIX
Frameworks: React, Rails, *Tauri*, *Flask*
Tools: Git, Docker, *tailscale*, MySQL, PostgreSQL
Other: Dean’s List · Bilingual — English and Mandarin